1. Project Title

Licenora

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2. Project Introduction

Licenora for SaaS is a platform designed to create, assign, validate, monitor, and manage software licenses for subscription-based products. It helps SaaS businesses control product access, manage customer entitlements, prevent unauthorized usage, and automate license lifecycle operations.

This system acts as a central layer between the SaaS product, billing logic, and customer accounts, ensuring that only authorized users or organizations can access features based on their active license plan.

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3. Problem the Project Solves

Many SaaS businesses struggle with:

- Manual license creation and tracking
- Inconsistent access control across plans
- Difficulty managing trial, active, expired, and suspended licenses
- Unauthorized account sharing or overuse
- Poor visibility into license utilization and renewals
- Lack of automation between subscriptions and product access

This project solves these problems by providing a structured system to issue and control licenses, track their status, and connect license rules directly to user access.

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4. Target Users

- SaaS founders and product owners
- B2B SaaS companies selling seat-based or plan-based subscriptions
- Internal operations and support teams
- Sales and account management teams
- Enterprise customers managing multiple user licenses
- Finance/admin teams monitoring renewals and entitlement status

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5. Key Features

- License key generation and assignment
- Plan-based license configuration
- Seat allocation and usage tracking
- License activation and validation API
- Trial, active, expired, suspended, and cancelled status handling
- Customer and organization-based license management
- Renewal and expiration alerts
- Feature entitlement control
- Admin dashboard for monitoring licenses
- Audit logs for license activity
- API/webhook support for billing and subscription sync
- Reports on active licenses, churn, renewals, and usage

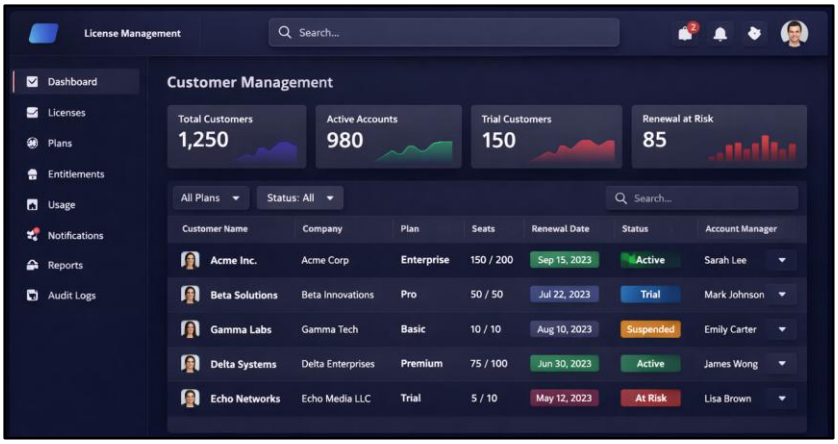
6. System Modules

1. Authentication and Access Control

- Admin login
- Role-based permissions
- Secure session management

2. Customer Management

- Create and manage customer accounts
- Store company and billing details
- Associate licenses with customers or organizations

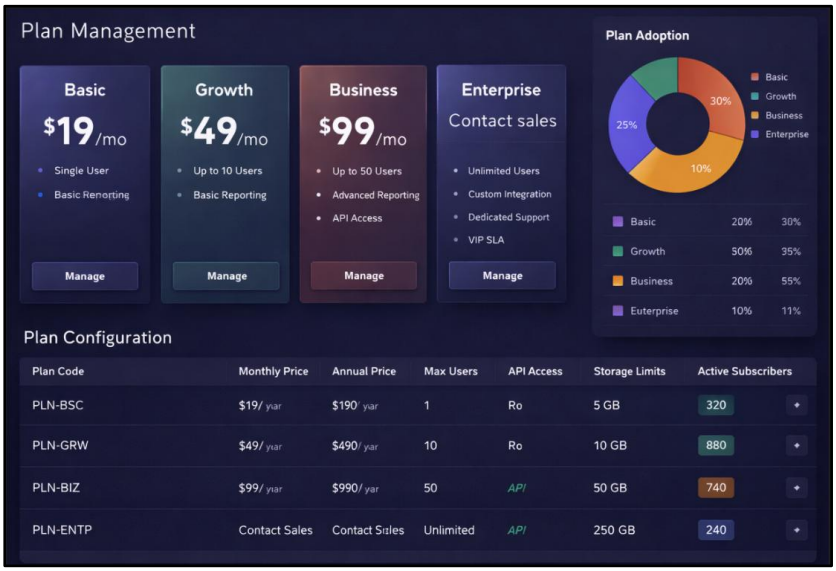


3. License Management

- Generate unique license keys
- Assign licenses to customers
- Activate, suspend, renew, revoke, or expire licenses
- Track license validity period

4. Subscription and Plan Management

- Define pricing plans
- Configure feature access by plan
- Set seat limits, duration, and renewal rules



5. Entitlement Management

- Map plans to product features
- Enable or disable features based on license type
- Handle usage-based restrictions

6. Validation API Module

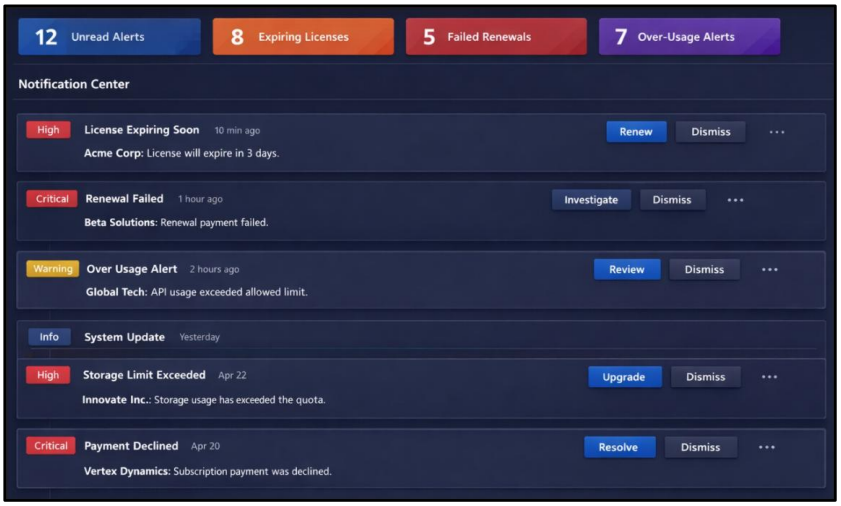
- API endpoint for real-time license verification
- Product-side validation requests
- Device/session/license usage checks

7. Usage Monitoring

- Track active seats and license consumption
- Detect overuse or abnormal activity
- Monitor license health and engagement

8. Notifications and Alerts

- Renewal reminders
- Expiry warnings
- Failed payment or suspension notifications
- Internal admin alerts

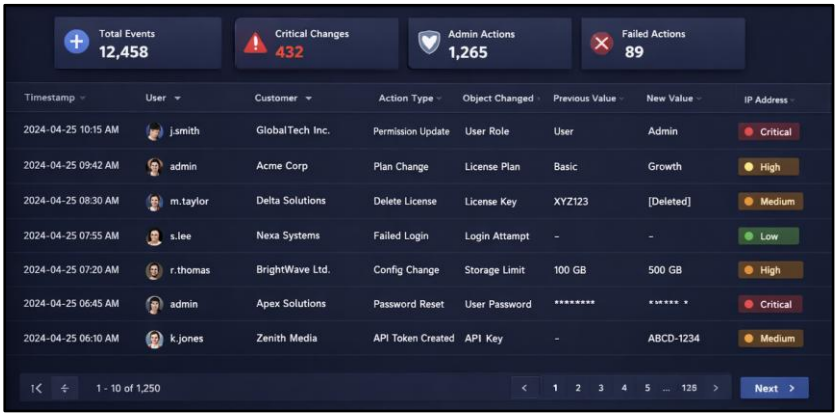


9. Reporting and Analytics

- Active vs expired licenses
- Revenue-linked license metrics
- License utilization reports
- Renewal and churn insights

10. Audit and Activity Logs

- Log license creation, assignment, updates, and status changes
- Maintain change history for compliance and troubleshooting



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7. User Roles

Super Admin

- Full system control
- Manage plans, licenses, customers, teams, and settings
- Access all reports and audit logs

Admin / Operations Manager

- Create and manage customer licenses
- Renew, suspend, or revoke licenses
- View operational reports

Support Agent

- View customer license details
- Assist with activation issues
- Resend license information and troubleshoot status problems

Customer Account Owner

- View purchased licenses
- Assign seats to team members
- Monitor renewal dates and usage

End User

- Access the SaaS product based on assigned entitlement
- Use only the features available under the active license

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8. Core Functional Requirements

- The system must allow admins to create software plans with configurable license rules.
- The system must generate unique licenses for each subscription or purchase.
- The system must assign licenses to individual users or organizations.
- The system must validate licenses in real time through an API.
- The system must support multiple license states such as trial, active, expired, suspended, and cancelled.
- The system must enforce access based on feature entitlements and seat limits.
- The system must track renewal and expiration dates.
- The system must notify customers and admins before license expiration.
- The system must maintain logs for all critical license events.
- The system must provide reporting dashboards for operational visibility.

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9. Admin Panel Features

- Dashboard with total licenses, active licenses, expiring licenses, and suspended licenses
- Customer and organization management
- Plan and pricing configuration
- License issuance and bulk assignment
- Manual renewal, extension, suspension, and revocation
- Seat usage monitoring
- Feature entitlement configuration
- Search and filter licenses by status, customer, plan, or date
- Notification template management
- Audit trail and activity viewer
- Export reports in CSV or Excel format
- API key and webhook configuration

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10. User Workflow / Journey

Admin Workflow

- Admin creates subscription plans and license rules.
- Customer purchases a SaaS subscription.
- System generates a license automatically or admin creates one manually.
- License is linked to the customer account or organization.
- Feature access and seat limits are applied based on the selected plan.
- Customer activates and uses the SaaS product.
- System validates the license during login or feature access.
- Before expiration, renewal reminders are sent.
- On renewal, the license is extended automatically or manually.
- On non-payment or cancellation, the license is suspended or expired.

Customer Workflow

- Customer signs up or is onboarded by sales/admin.
- Customer receives an active license or subscription confirmation.
- Customer logs into the SaaS product.
- Access is granted based on active license entitlements.
- Team owner assigns seats to internal users if applicable.
- Customer receives alerts for upcoming expiration or renewal.
- Customer upgrades, renews, or cancels subscription as needed.

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11. Tech Stack

Backend

- ASP.NET Core Web API

Frontend

- Angular
- Tailwind CSS
- PrimeNG

Database

- MS SQL Server

Cloud / DevOps

- Azure
- Azure DevOps